QP Code: 25900284

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION October 2025
Third SEMESTER
Discipline Specific Core Course (DSC) - MG3DSCCHE201 - ORGANIC CHEMISTRY - I
(2024 ADMISSION ONWARDS)

Instruction to the Examiners

- *This scheme is only a guideline for valuation. Any other attempt which is equally valid and correct may be given due marks.*
- *Exceptions are there in some cases and if the answer is complete full marks may be given even if the required number of valuation points could not be identified.*
- *Some of the answers given in the scheme are elaborate and this is to help the examiner in their valuation. Please do not expect all those points in the answers of the students.*

Part A

Multiple choice questions. Answer all questions Each question carries 1 mark

1. The general formula of alkyne is:
c). C_nH_{2n-2}

2. Fluorination of benzene is rarely carried out directly because:
c) The reaction is uncontrollably violent

3. Choose a type of configurational isomerism from below.
(a) Optical isomerism

4. The catalyst commonly used in hydrogenation is:
(c) Pt/Pd/Ni

5. Which description best connects benzene's resonance energy with its molecular orbital picture?
(b) Resonance energy arises from alternating Kekulé structures

6. According to the Sequence rule in Cahn-Ingold-Prelog (CIP) system, which among the following has the lowest priority?
(d) $–CH_2CH_2CH_2CH_3$

7. Acetylenes can be regenerated from their metal acetylides on treating with
(b) H_2O

8. Which of the following compounds is not aromatic?



c)

9. A molecule is said to be chiral:
(a) If it cannot be superimposed on its mirror image

10. Alkenes show:
(b) Both sigma and pi bonds

Part B

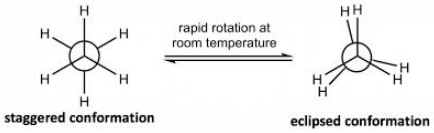
Short Answer Questions, Answer any 4 questions, Each question carries 3 marks

11. Describe the addition of bromine to an alkyne. What products are obtained?

- Alkyne + Br_2 → trans-dibromoalkene ***(1 mark)***
- trans-dibromoalkene → tetrahaloalkane on excess Br_2. ***(1 mark)***
- Anti-addition mechanism. ***(1 mark)***

12. Toluene undergoes electrophilic aromatic substitution about 4000 times faster than benzene. Explain the reason.
$–CH_3$ is an activating group towards electrophilic substitution reaction [electron donating -+I effect]. The electron donating effect of alkyl group tend to decrease the positive charge on the carbocation and thus stabilize the ion and increase the rate of the substitution.
(Any two valid points – 3 marks)

13. Which is the most stable conformer of ethane? Why?
The staggered conformer of ethane is the most stable because it minimizes electron repulsion between adjacent C–H bonds, providing the lowest torsional strain.



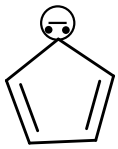
(Any two valid points with valid structures – 3 marks)

14. Why is peroxide effect observed only with HBr?

- Follows free radical mechanism.
- HCl and HI form unstable or non-feasible radicals.
- Only HBr gives anti-Markovnikov addition.

(Any valid three points – 3 marks)

15. Discuss the aromatic character of the cyclopentadienyl anion.



Follows Huckel's rule → Cyclic, planar, cyclic π electron cloud; 2 π bonds + 2 electrons = 6 π electrons; 4n+ 2 = 6; n =1; therefore aromatic.

(Structure – 1 mark; Any two valid points describing the aromaticity - 2 marks)

16. Give a brief account of any three physical properties of alkanes.

- Non-polar; insoluble in water.
- Density < 1 g/cm³.
- Boiling point increases with chain length.

(Any three valid points– 3 marks)

Part C Short Essay Questions

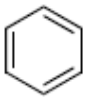
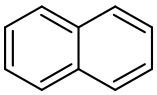
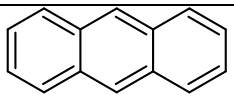
Answer any 2 questions; Each question carries 7 marks

17. Explain the free radical substitution mechanism of methane chlorination with all steps.

- **Initiation:** $Cl_2 \rightarrow 2Cl\bullet$ (by light/heat).
- **Propagation:**
  $CH_4 + Cl\bullet \rightarrow CH_3\bullet + HCl$
  $CH_3\bullet + Cl_2 \rightarrow CH_3Cl + Cl\bullet$
- **Termination:** Combination of radicals.
- Product: CH_3Cl (and higher chlorinated products).

(Correct mechanism with product – 7 marks; (For each steps give – 3 x 2=6 marks and for the product – 1 mark))

18. Compare the structures and aromatic character of benzene, naphthalene, and anthracene.

Property	Benzene	Naphthalene	Anthracene
Structure			
No. of rings	1	2 (fused)	3 (linearly fused)
π-Electrons	6	10	14
Hückel's rule	n = 1	n = 2	n = 3
Resonance energy (approx.)	151 kJ/mol	255 kJ/mol	351 kJ/mol

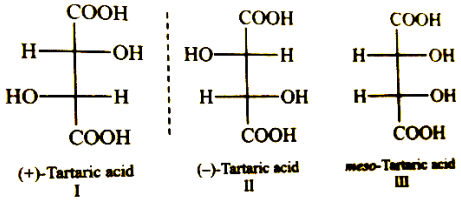
Aromatic stability order: Benzene > Naphthalene > Anthracene

(As ring fusion increases, delocalization per ring decreases due to π-electron sharing.)

(Three structure – 3 marks, Any two valid points including* Hückel*'s rule – 4 marks)

19. Draw all possible stereoisomers of tartaric acid. Identify which are enantiomers, diastereomers, and meso compounds. Explain your reasoning.

Tartaric acid (HOOC–CH(OH)–CH(OH)–COOH) has two chiral centers. D-(+)-tartaric acid and L-(–)-tartaric acid are enantiomers and are optically active. Meso-tartaric acid is optically inactive due to internal symmetry. Enantiomers are mirror images; the meso form is a diastereomer of both.



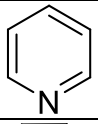
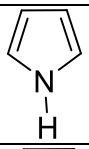
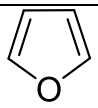
(Correct structures – 4 marks; Any three valid points for reasoning - 3 marks)

Part D Essay Type Questions

Answer any 1 question; Each question carries 14 marks

20.Compare the aromaticity in heterocycles: pyridine, pyrrole, and furan. Comment on their basicity differences.

- Pyridine, pyrrole, and furan are **heteroaromatic compounds**
- Each satisfies **Hückel's 4n + 2 π-electron rule** (n = 1).

Compound	Structure	Aromaticity Explanation
Pyridine		Cyclic, Planar, fully conjugated ring with 6 π-electrons (from three C=C bonds); nitrogen's lone pair lies in an sp^2 orbital **outside** the π-system → **aromatic**.
Pyrrole		Cyclic, Planar, conjugated five-membered ring with 4 π-electrons from two C=C bonds; nitrogen **contributes its lone pair** to the π-system, making a total of 6 π-electrons → **aromatic**.
Furan		Cyclic, Planar, conjugated five-membered ring with 4 π-electrons from two C=C bonds; oxygen **donates one of its lone pairs** into conjugation, giving 6 π-electrons → **aromatic**, though **less so** than pyrrole or pyridine due to oxygen's high electronegativity.

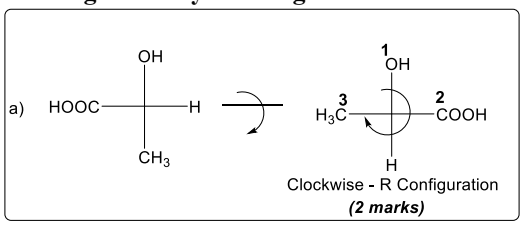
Aromatic stability order:

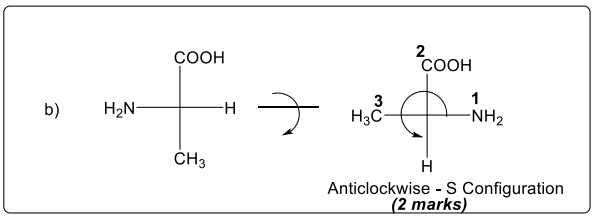
Pyridine > Pyrrole > Furan (based on resonance energy and delocalization efficiency).

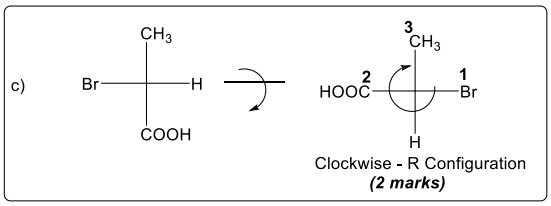
(Structure of three compounds – 1 x 3= 3 marks, Explanation on aromaticity including* Hückel*'s rule (Any two valid points)→ 3x3=9 marks;

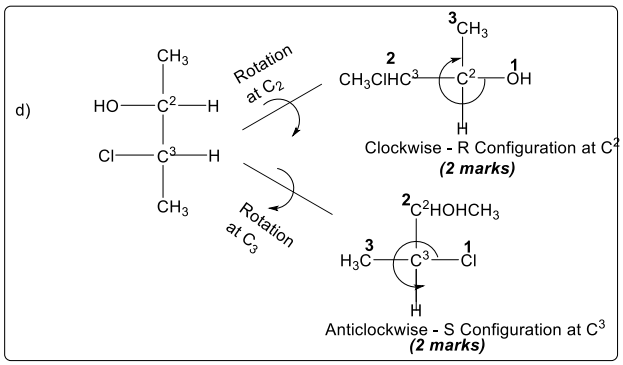
Second part of the question is out of syllabus→ Give 2 marks to all)

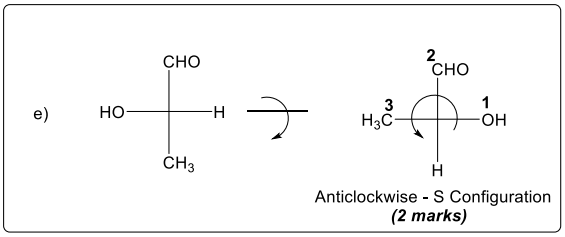
21. Assign the R and S configuration to each of the chiral centers in the following molecules. Show the reasoning used in your assignments

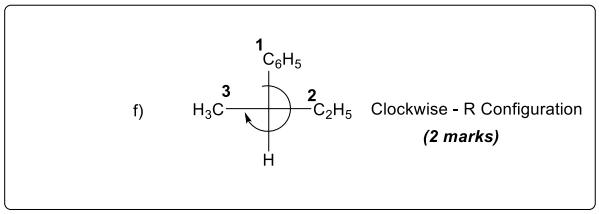












(Answers a-c,e,f – 2 marks each; d – 4 marks)
